1. $P(A) = \dfrac{\text{Số trường hợp thuận lợi cho } A}{\text{Số trường hợp có thể của phép thử}}$ **2.** $P(A) = \dfrac{m(A)}{m(\Omega)}$

3. $P(A \cup B) = P(A) + P(B) - P(AB)$

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$

$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_i P(A_i) - \sum_{i<j} P\left(A_iA_j\right) + \sum_{i<j<k} P\left(A_iA_jA_k\right) - \ldots + (-1)^{k-1} \sum_{i_1<i_2<\ldots<i_k} P\left(A_{i_1}A_{i_2}\ldots A_{i_k}\right) + \ldots + (-1)^{n-1} P(A_1A_2\ldots A_n)$

4. $P(A_1 \cup A_2 \cup \ldots \cup A_n) \stackrel{?}{=} P(A_1) + P(A_2) + \ldots + P(A_n)$

5. $P(A) + P\left(\overline{A}\right) = 1$ **6.** $P(A \mid B) = \dfrac{P(AB)}{P(B)}$

7. $P(A \mid B) = \dfrac{\text{Số trường hợp thuận lợi cho } A \text{ khi } B \text{ đã xảy ra}}{\text{Số trường hợp có thể của phép thử (tạo ra } A\text{) khi } B \text{ đã xảy ra}}$

8. $P(A_1A_2\ldots A_n) = P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_1A_2) \cdots P(A_n \mid A_1A_2\ldots A_{n-1})$

9. $P(AB) \stackrel{?}{=} P(A) \cdot P(B)$

10. $P(B) \stackrel{?}{=} \sum_{i=1}^{n} P(A_i) \cdot P(B \mid A_i)$ **11.** $P(A_i \mid B) = \dfrac{P(A_i) \cdot P(B \mid A_i)}{P(B)}$

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12. $f(x) = P(X = x)$ **13.** $\sum_x f(x) = 1, \quad \int_{-\infty}^{\infty} f(x)\,dx = 1$

14. $P(X \in A) = \sum_{x \in A} f(x), \quad P(a \leq X \leq b) = \int_a^b f(x)\,dx$

15. $F(x) = P(X \leq x) = \begin{cases} \sum_{t \leq x} f(t) \\ \int_{-\infty}^{x} f(t)\,dt \end{cases}$

16. $F'(x) \stackrel{?}{=} f(x)$
$F(-\infty) = 0, \; F(\infty) = 1$
$P(a < X \leq b) = F(b) - F(a)$

17. $Y = \varphi(X)$: $\quad \mathrm{Im}Y = \varphi(\mathrm{Im}\,X) = \{y = \varphi(x) \mid x \in \mathrm{Im}X\}; \quad P(Y = y) = P[\varphi(X) = y] = \sum_{\varphi(x)=y} f(x)$

18. $Y = \varphi(X)$: $\quad \varphi(x) < y \Leftrightarrow a(y) < x < b((y); \quad g\,(y) = b'(y) f[b((y)] - a'(y) f[a(y)]$

19. $EX = \begin{cases} \sum_x xf(x) \\ \int_{-\infty}^{\infty} xf(x)\,dx \end{cases} : \quad E(X + Y) = EX + EY, \quad E(kX) = k \cdot EX$

20. $E[\varphi(X)] = \begin{cases} \sum_x \varphi(x)\, f(x) \\ \int_{-\infty}^{\infty} \varphi(x) f(x)\,dx \end{cases}$ **21.** $E\left(X^2\right) = \begin{cases} \sum_x x^2 f(x) \\ \int_{-\infty}^{\infty} x^2 f(x)\,dx \end{cases}$

22. $\mathrm{Var}(X) = E\left(X^2\right) - (EX)^2$: $\quad \mathrm{Var}(kX) = k^2\,\mathrm{Var}(X), \quad \mathrm{Var}(X + Y) \stackrel{?}{=} \mathrm{Var}(X) + \mathrm{Var}(Y)$

23. $\sigma_X = \sqrt{\mathrm{Var}(X)}$

24. $X \sim B(n,p)$: $f(x) = C_n^x p^x (1-p)^{n-x}, x = 0,1,\ldots,n$; $EX = np$, $\mathrm{Var}(X) = np(1-p)$

25. $X \sim \mathrm{Poisson}(\lambda)$: $f(x) = e^{-\lambda}\dfrac{\lambda^x}{x!}, x = 0,1,2,\ldots$; $EX = \lambda$, $\mathrm{Var}(X) = \lambda$

26. $X \sim U(a,b)$: $f(x) = \begin{cases} \dfrac{1}{b-a}, & x \in [a,b] \\ 0, & x \notin [a,b] \end{cases}$; $EX = \dfrac{a+b}{2}$, $\mathrm{Var}(X) = \dfrac{(b-a)^2}{12}$

27. $X \sim \mathrm{Exp}(\lambda)$: $f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & x \le 0 \end{cases}$; $EX = \dfrac{1}{\lambda}$, $\mathrm{Var}(X) = \dfrac{1}{\lambda^2}$

28. $X \sim N(\mu, \sigma^2)$: $f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$; $EX = \mu$, $\mathrm{Var}(X) = \sigma^2$; $Z = \dfrac{X-\mu}{\sigma} \sim N(0,1)$

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29. $f(x,y) = P(X = x, Y = y)$

30. $\sum\limits_{x,y} f(x,y) = 1$, $\iint_{\mathbb{R}^2} f(x,y)\,dxdy = 1$

31. $P[(X,Y) \in A] = \begin{cases} \sum\limits_{(x,y)\in A} f(x,y) \\ \iint_A f(x,y)\,dxdy \overset{?}{=} \int_a^b dx \int_{c(x)}^{d(x)} f(x,y)\,dy \end{cases}$

32. $F(x,y) = P(X \le x, Y \le y) = \begin{cases} \sum\limits_{u\le x, v\le y} f(u,v) \\ \iint\limits_{u\le x, v\le y} f(u,v)\,dudv \end{cases}$

33. $\dfrac{\partial^2}{\partial x \partial y} F(x,y) \overset{?}{=} f(x,y)$

34. $f_1(x) = \begin{cases} P(X = x) = \sum\limits_y f(x,y) \\ \int_{-\infty}^{\infty} f(x,y)\,dy \end{cases}$, $f_2(y) = \begin{cases} P(Y = y) = \sum\limits_x f(x,y) \\ \int_{-\infty}^{\infty} f(x,y)\,dx \end{cases}$

35. $f(x,y) \overset{?}{=} f_1(x) \cdot f_2(y)\ \ \forall x,y$

36. $\varphi(x \mid Y = y) = \left[P(X = x \mid Y = y) = \right] \dfrac{f(x,y)}{f_2(y)}\ \forall x$

37. $Z = X + Y$: $P(Z = z) = P(X + Y = z) = \sum\limits_{x+y=z} f(x,y)$; $g(z) = \int_{-\infty}^{\infty} f(x, z-x)\,dx$

38. $E[\varphi(X,Y)] = \begin{cases} \sum\limits_{x,y} \varphi(x,y) \cdot f(x,y) \\ \iint_{\mathbb{R}^2} \varphi(x,y) \cdot f(x,y)\,dxdy \end{cases}$

39. $E(XY) = \begin{cases} \sum\limits_{x,y} xyf(x,y) \\ \iint_{\mathbb{R}^2} xyf(x,y)\,dxdy \end{cases}$

40. $\mathrm{Cov}(X,Y) = E(XY) - EX \cdot EY$

41. $\rho_{XY} = \dfrac{\mathrm{Cov}(X,Y)}{\sigma_X \cdot \sigma_Y}$

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42. $P(|X - EX| \geqslant \varepsilon) \leqslant \dfrac{DX}{\varepsilon^2}$

43. $\overline{X} \xrightarrow{p} a, \quad \dfrac{m}{n} \xrightarrow{p} p, \quad \dfrac{\overline{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$

44. $X \sim B(n,p),\ n \gg 1: \quad P(k_1 \leqslant X \leqslant k_2) \approx \Phi\left[\dfrac{k_2 - np}{\sqrt{np(1-p)}}\right] - \Phi\left[\dfrac{k_1 - np}{\sqrt{np(1-p)}}\right]$

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45. $\overline{X} = \dfrac{1}{n}\sum_{i=1}^{n} X_i$

46. $\overline{x} = \dfrac{1}{n}\sum_{i=1}^{n} x_i = \dfrac{1}{n}\sum_{i=1}^{k} m_i x_i$

47. $S_n^2 = \dfrac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2 = \left(\dfrac{1}{n}\sum_{i=1}^{n} X_i^2\right) - \overline{X}^2$

48. $s_n^2 = \left(\dfrac{1}{n}\sum_{i=1}^{n} x_i^2\right) - \overline{x}^2 = \left(\dfrac{1}{n}\sum_{i=1}^{k} m_i x_i^2\right) - \overline{x}^2$

49. $S^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2 = \dfrac{n}{n-1} S_n^2$

50. $s^2 = \dfrac{1}{n-1}\left[\left(\sum_{i=1}^{n} x_i^2\right) - n\overline{x}^2\right] = \dfrac{1}{n-1}\left[\left(\sum_{i=1}^{k} m_i x_i^2\right) - n\overline{x}^2\right]$

51. $P(Z > z_\alpha) = \alpha, \qquad P\left(\chi_k^2 > \chi_{\alpha,k}^2\right) = \alpha, \qquad P\left(t_k > t_{\alpha,k}\right) = \alpha$

52.

Tham số	*Kiểu ước lượng*			*Ghi chú*
	ƯLKC	*Khoảng tin cậy với độ tin cậy* $1-\alpha$		
EX	$\overline{X}$	$DX = \sigma^2$ đã biết	$\left(\overline{X} - z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}, \overline{X} + z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}\right)$	với KTC: $X \sim N$
		DX chưa biết	$\left(\overline{X} - t_{\alpha/2,n-1}\dfrac{S}{\sqrt{n}}, \overline{X} + t_{\alpha/2,n-1}\dfrac{S}{\sqrt{n}}\right)$	
DX	S^2	$\left(\dfrac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \dfrac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right)$		
$p = P(A)$	$p^* = \dfrac{m}{n}$	$\left(p^* - z_{\alpha/2}\sqrt{\dfrac{p^*(1-p^*)}{n}}, p^* + z_{\alpha/2}\sqrt{\dfrac{p^*(1-p^*)}{n}}\right)$		$n \gg 1$

53.

H_0	H_1	*Dấu hiệu / ĐK*	*Thống kê kiểm định*	*ĐK bác bỏ* H_0
$EX = \mu_0$	$EX \neq \mu_0$	$DX = \sigma^2$ đã biết	$z = \dfrac{\overline{x} - \mu_0}{\sigma}\sqrt{n}\ (*)$	$\lvert z\rvert > z_{\alpha/2}$
	$EX > \mu_0$			$z > z_\alpha$
	$EX < \mu_0$			$z < -z_\alpha$
như trên		DX chưa biết	$t = \dfrac{\overline{x} - \mu_0}{s}\sqrt{n}\ (**)$	$\lvert t\rvert > t_{\alpha/2,n-1}$
				$t > t_{\alpha,n-1}$
				$t < -t_{\alpha,n-1}$
$p = p_0$	$p \neq p_0$	$n \gg 1$	$z = \dfrac{\frac{m}{n} - p_0}{\sqrt{p_0(1-p_0)}}\sqrt{n}$	như $(*)$
	$p > p_0$			
	$p < p_0$			
$EX_1 = EX_2$	$EX_1 \neq EX_2$	DX_1, DX_2 đã biết	$z = \dfrac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{DX}{n_1} + \frac{DY}{n_2}}}$	như $(*)$
	$EX_1 > EX_2$			

	$EX_1 < EX_2$			
như trên		$DX_1 = DX_2$ chưa biết	$t = (1)$	như $(**)$, thay $n-1$ bởi n_1+n_2-2
$p_1 = p_2$	$p_1 \neq p_2$ $p_1 > p_2$ $p_1 < p_2$	$n_1, n_2 \gg 1$	$z = (2)$	như $(*)$

$$(1) = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{(n_1-1)s_{X_1}^2 + (n_2-1)s_{X_2}^2}} \sqrt{\frac{n_1 n_2 (n_1+n_2-2)}{n_1+n_2}}$$

$$(2) = \left(\frac{m_1}{n_1} - \frac{m_2}{n_2}\right) \sqrt{\frac{n_1 n_2 (n_1+n_2)}{(m_1+m_2)(n_1+n_2-m_1-m_2)}}.$$

54. $H_0 : P(A_i) = p_{i0}\ \forall i = \overline{1,k}$: **(a)** $e_i \stackrel{?}{=} np_{i0}$, **(b)** $\chi^2 = \sum_{i=1}^{k} \frac{(o_i - e_i)^2}{e_i}$, **(c)** $\chi^2 \stackrel{?}{>} \chi^2_{\alpha,k-1}$

55. $H_0 : X \sim F(x)$: **(a)** $\mathrm{Im}X = S_1 \cup S_2 \cup \ldots \cup S_k$, **(b)** $p_{i0} = P(X \in S_i \mid H_0) = \begin{cases} \sum\limits_{x \in S_i} f(x) \\ \int\limits_{S_i} f(x)\,dx \end{cases}$, **(54)**

56. $H_0 : X \sim F(x, \theta_1, \theta_2, \ldots, \theta_r) \Leftrightarrow H_0^* : X \sim F\left(x, \theta_1^*, \theta_2^*, \ldots, \theta_r^*\right)$, $\chi^2 \stackrel{?}{>} \chi^2_{\alpha,k-r-1}$:

$X \sim ?$	θ	θ^*
$B(n,p)$	p	p^*
Poisson(λ)	λ	$\overline{X}$
$U(a,b)$	a, b	$\overline{X} - S_n\sqrt{3},\ \overline{X} + S_n\sqrt{3}$
Exp(λ)	λ	$1/\overline{X}$
$N\left(\mu, \sigma^2\right)$	μ, σ^2	$\overline{X}, S_n^2$

57. $H_0 : X, Y$ độc lập: **(a)** $\mathrm{Im}X = S_1 \cup \ldots \cup S_r,\ \mathrm{Im}Y = T_1 \cup \ldots \cup T_c$; **(b)** $n_{i*} = \sum_j n_{ij},\ n_{*j} = \sum_i n_{ij}$;

(c) $e_{ij} = \frac{n_{i*} \times n_{*j}}{n}$; **(d)** $\chi^2 = \sum_{i,j} \frac{\left(n_{ij} - e_{ij}\right)^2}{e_{ij}}$; **(e)** $\chi^2 \stackrel{?}{>} \chi^2_{\alpha,(r-1)(c-1)}$

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58. $r = \dfrac{\overline{XY} - \overline{X} \cdot \overline{Y}}{S_{n,X} S_{n,Y}}$

59. $Z = E(Y \mid X)$: $X = x \Rightarrow Z = E(Y \mid X = x) = \begin{cases} \frac{1}{f_1(x)} \sum\limits_y y f(x,y) \\ \frac{1}{f_1(x)} \int_{-\infty}^{\infty} y f(x,y) dy \end{cases}$

60. $E\left\{\left[Y - b_1 X - b_0\right]^2\right\} \to \min\limits_{b_1, b_0}$: $b_1 = \rho \dfrac{\sigma_Y}{\sigma_X}, b_0 = EY - b_1 EX$

61. $\dfrac{1}{n} \sum\limits_{i=1}^{n} \left(Y_i - b_1 X_i - b_0\right)^2 \to \min\limits_{b_1, b_0}$: $b_1 = r \dfrac{S_Y}{S_X}, b_0 = \overline{Y} - b_1 \overline{X}$